

Multilingual Emotion Neurons in Large Audio-Language Models

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Abstract

Emotion is central to human communication, and its expression varies across languages. Large audio-language models (LALMs) achieve strong performance on multilingual speech tasks, yet it remains unclear whether they encode emotion through language-specific correlations or language-agnostic representations. We present the first neuron-level interpretability study of this question. We define Multilingual Emotion Neurons (MLENs) as functional units exhibiting stable emotional selectivity and aligned causal effects across languages, and introduce Consistency-Regularized Fusion (CR-Fusion) to identify them. Across four modern LALMs and 12 typologically diverse languages, emotion-sensitive neurons identified independently per language show minimal overlap, and additional monolingual identification data saturates quickly without isolating more transferable units, motivating identification from pooled cross-lingual evidence. Causal interventions demonstrate that MLENs identified by CR-Fusion provide more precise and transferable affective control than monolingual neuron sets in both zero-shot and low-resource settings. Leave-one-out ablations further reveal asymmetric transfer: individual identification languages, including low-resource ones, contribute non-redundant evidence, while several low-resource languages benefit most from the resulting cross-lingual transfer. Together, our findings provide the first causal, neuron-level account of how LALMs encode emotion across languages, and establish multilingual neuron identification as an effective mechanism for understanding cross-lingual affective behavior.

1 Introduction

Emotion is a core channel of human communication: beyond what we say, speech conveys how we feel through prosody, timing, intensity, and culturally shaped display rules (Wilce, 2009; Lindquist

et al., 2015). In a multilingual world, this affective layer must be robust to linguistic diversity: people routinely infer emotion from unfamiliar languages, yet recognition is modulated by language, culture, and learned acoustic conventions (Russell, 1991; Pell et al., 2009; Jackson et al., 2019). This tension between pan-cultural cues and language-specific realizations makes multilingual emotion an ideal testbed for probing the generalizability of concept representations in modern foundation models.

Decades of cross-lingual SER research show that emotion cues partially generalize to unseen languages, yet transfer remains sensitive to language and corpus mismatch, and human and model transfer patterns diverge systematically (Albornoz and Milone, 2017; Han et al., 2025). It thus remains open to what extent current models rely on cross-lingually shared affective representations versus dataset-, corpus-, and language-local correlations.

Parallel interpretability work shows that individual neurons in LLMs and multimodal models align with human-interpretable concepts, including affective states (Huben et al., 2024; Bau et al., 2017; Sofroniew et al., 2026). However, emotion-correlated neurons identified in one language often fail to generalize to others in multilingual encoders (Singh et al., 2026), and cross-lingual generalization of activation steering remains contested (Maraia et al., 2026). This tension between language-specific and language-agnostic emotion representations motivates our investigation (§2).

Modern multimodal foundation models (Yao et al., 2024; Xu et al., 2025b), particularly LALMs (KimiTeam, 2025; Goel et al., 2025), offer a new paradigm for probing this question. By integrating vast acoustic and linguistic pre-training, LALMs have achieved competitive performance on a broad range of spoken language understanding tasks (Huang et al., 2024a; Sakshi et al., 2025), including speech emotion recognition (Cheng et al., 2024; He et al., 2025). However, mechanistic ac-

counts of how these models represent affective information remain lacking. It is unclear whether LALMs utilize language-specific functional units or possess a shared representational structure that generalizes emotion processing across broad linguistic diversity.

To the best of our knowledge, this work represents the first systematic attempt to investigate and manipulate the intrinsic multilingual emotion representations within LALMs. Using SER as a probe task, we identify affective neurons through activations across four open-source LALMs, Audio-Flamingo-3 (Goel et al., 2025), Qwen2.5-Omni-7B (Xu et al., 2025b), MiniCPM-o-4.5 (Yao et al., 2024), and Kimi-Audio (KimiTeam, 2025), and 12 typologically diverse languages, including high-resource languages (e.g., English, Mandarin) and low-resource languages such as Amharic, Bengali and Urdu, where emotional speech data is often sparse (Koehn and Knowles, 2017; Mohmad Dar and Delhibabu, 2024). To isolate neurons with stable cross-lingual emotional selectivity, we propose *Consistency-Regularized Fusion* (CR-Fusion) that operates atop any neuron selector (e.g., Mean-activation-based (Bau et al., 2019; Dalvi et al., 2019)) to leverage multilingual evidence for identifying cross-lingually stable affective units.

Our investigation yields five principal findings. (1) **Emotion-sensitive neurons (ESNs) identified independently in different languages exhibit minimal set intersection** with weak to moderate rank correlation in selectivity scores, revealing that monolingual identification procedures select substantially different neurons even when underlying ranking structures are similar. (2) **Monolingual evidence saturates rapidly** beyond 50 instances; additional within-language data yields no further improvement in isolating transferable neurons. (3) We define Multilingual Emotion Neurons (MLENs) as functional units exhibiting stable emotional selectivity across languages. Causal interventions confirm that **MLENs play a functional role in emotion processing**: deactivation selectively impairs recognition of target emotions, while steering selectively enhances it. Our proposed CR-Fusion **matches or outperforms the best monolingual identification** in zero-shot and low-resource settings across models, with the exception of steering on Qwen2.5-Omni-7B, the model with the lowest language-invariant share. (4) Causal effects are heterogeneous across emotions: anger, happiness and sadness show the largest and most stable inter-

vention effects, while fear and neutral are weaker and more variable. This ordering tracks baseline recognition accuracy and class support rather than representational universality—per-emotion invariant shares are in fact highest for neutral and fear. (5) We uncover **asymmetric transfer patterns**, with individual identification languages contributing non-substitutable evidence. These results demonstrate that **low-resource languages can contribute non-redundant evidence for multilingual neuron identification while simultaneously benefiting from improved cross-lingual transfer**.

These findings provide a mechanistic framework for understanding how multimodal foundation models process emotion across languages, with practical implications for training-free cross-lingual generalization and equitable affective computing for low-resource language communities.

2 Related Work

Multilingual Speech Emotion Recognition. Research on multilingual SER has progressed from early cross-lingual transfer using traditional deep neural networks (Albornoz and Milone, 2017; Neumann and Thang Vu, 2018; Zehra et al., 2021) to more scalable approaches leveraging self-supervised encoders such as wav2vec 2.0 and transformer-based architectures (Sharma, 2022; Al-onazi et al., 2022). Recent benchmarks like EmoBox (Ma et al., 2024) have standardized multilingual evaluation while highlighting persistent cross-lingual robustness challenges, with studies revealing systematic divergences between human and model transfer patterns (Singh and Gupta, 2023; Han et al., 2025). Current work explores LALMs for zero-shot cross-lingual recognition, emotion captioning, and joint audio-text reasoning (Zou et al., 2025; Xu et al., 2024; He et al., 2025), complemented by parameter-efficient methods like LoRA for cross-lingual alignment (Goncalves et al., 2024). However, there is no mechanistic study of emotion representation across languages in LALMs (Shou et al., 2025).

Neuron-Level Concept and Emotion Interpretability in Audio-Language Models. Identifying functional units that respond selectively to human-interpretable concepts is a long-standing theme in interpretability, established in vision (Bau et al., 2017, 2020) and LLMs (Huben et al., 2024; Voita et al., 2024; Yu and Ananiadou, 2024; Tang et al., 2024). For affect specifically, recent work

identifies clustered emotion neurons in LLMs (Lee et al., 2025) that are causally controllable via activation steering (Wang et al., 2025), with Sofroniew et al. (2026) showing how affective states decompose into modular circuits; yet Singh et al. (2026) find that emotion-correlated neurons identified in one language fail to generalize in multilingual encoders, and cross-lingual generalization of steering remains contested (Maraia et al., 2026). In LALMs, neuron-level analysis has so far targeted modality attribution and generic acoustic concepts (Huo et al., 2024; Wu et al., 2024), or probed phonetic and prosodic encoding layer-wise (Yang et al., 2025); affective units have been identified only monolingually, for SER (Zhao et al., 2026b) and emotional voice conversion (Zhao et al., 2026c). Their stability across linguistic boundaries has not been explored, which motivates our investigation (extended discussion in Appendix C).

3 Method

Our pipeline comprises three stages: (1) activation logging on correctly recognized SER instances, (2) neuron scoring and selection using principled fusion strategies that integrate language-conditioned statistics into cross-lingual neuron sets, and (3) causal analysis through targeted intervention.

3.1 Multilingual Activation Logging

Let $\mathcal{L}$ denote a set of languages and $\mathcal{E}$ the set of emotions. We run the unintervened model on an SER task of emotion-labeled speech and restrict logging to correctly predicted items to reduce contamination from failure modes and to obtain cleaner emotion-conditioned statistics. Let $\mathcal{D}^{(\ell,e)}$ denote the set of correctly predicted utterances of emotion $e \in \mathcal{E}$ in language $\ell \in \mathcal{L}$. For an utterance x with token positions $t \in \{1, \dots, T_x\}$, let $a_{l,n,t}(x)$ denote the scalar SwiGLU gate activation (Shazeer, 2020) of neuron n in layer l , and let $w_{x,t} \in \{0, 1\}$ mask out irrelevant positions (e.g., padding and instruction-prompt tokens). We aggregate positive-activation counts and valid token counts as

$$K_{l,n}^{(\ell,e)} = \sum_{x \in \mathcal{D}^{(\ell,e)}} \sum_{t=1}^{T_x} w_{x,t} \mathbb{I}[a_{l,n,t}(x) > 0],$$

$$T^{(\ell,e)} = \sum_{x \in \mathcal{D}^{(\ell,e)}} \sum_{t=1}^{T_x} w_{x,t},$$
(1)

and form the activation-probability profile $P_{l,n}^{(\ell,e)} = K_{l,n}^{(\ell,e)} / T^{(\ell,e)}$.

3.2 Monolingual Evidence and Neuron Identification Methods

We first obtain monolingual evidence by scoring neurons independently per language. Let $s_{l,n}^{(\ell,e)}$ denote a selector score derived from $P_{l,n}^{(\ell,e)}$. We use a margin-based selector, **Contrastive Activation Margin (ConAct)** (Zhao et al., 2026a), as our default due to its consistently stronger intervention effectiveness in comparison with LAP (Huben et al., 2024; Gurnee et al., 2024), LAPE (Tang et al., 2024; Namazifard and Poech, 2025), and MAD (Bau et al., 2019; Dalvi et al., 2019). ConAct assigns each neuron to its most preferred emotion and measures a margin against the second most preferred. Using $P_{l,n}^{(\ell,e)}$, we define: $P_{l,n}^{(1)}(\ell) = \max_{e \in \mathcal{E}} P_{l,n}^{(\ell,e)}$

$$e_{l,n}^{(1)}(\ell) = \arg \max_e P_{l,n}^{(\ell,e)}, P_{l,n}^{(2)}(\ell) = \max_{e \neq e_{l,n}^{(1)}(\ell)} P_{l,n}^{(\ell,e)}$$

$$s_{l,n}^{(\ell,e)} = \begin{cases} P_{l,n}^{(1)}(\ell) - P_{l,n}^{(2)}(\ell), & e = e_{l,n}^{(1)}(\ell) \\ 0, & \text{otherwise} \end{cases}.$$

For each (ℓ, e) , we rank neurons by the emotion-conditioned score $s_{l,n}^{(\ell,e)}$ and select a fixed fraction $r=0.5\%$ to obtain monolingual ESN sets $\mathcal{I}_l^{(\ell,e)}$.

3.3 Fusion Strategies for Multilingual Emotion Neurons

We now construct MLENs by fusing monolingual evidence $\{s_{l,n}^{(\ell,e)}\}_{\ell \in \mathcal{L}}$. The goal is to identify neurons that are not only emotion-selective, but also consistent across languages. Our designs are inspired by multi-view learning and ensembling principles (e.g., intersection/consensus, averaging, and variance-regularized objectives), adapted here to neuron selection.

Overlap Fusion. This conservative strategy retains neurons supported by *all* languages. We form a fixed-budget MLEN set by re-ranking candidates with a joint score that prefers agreement (i.e. by its weakest per-language evidence), $s_{l,n}^{\text{overlap},(e)} = \min_{\ell \in \mathcal{L}} \tilde{s}_{l,n}^{(\ell,e)}$, where $\tilde{s}$ denotes per-language normalized scores (z-score or robust z-score across neurons). We then select the top- r neurons by s^{overlap} subject to a same-emotion constraint (the preferred emotion agrees across languages).

Joint Fusion. This strategy treats all languages as a single unified dataset by pooling raw activation statistics before computing selector scores. We first merge the activation counts and token counts across all languages: $K_{l,n}^{(\text{joint},e)} = \sum_{\ell \in \mathcal{L}} K_{l,n}^{(\ell,e)}$, $T^{(\text{joint},e)} = \sum_{\ell \in \mathcal{L}} T^{(\ell,e)}$. Then, we compute the pooled activation probability: $P_{l,n}^{(\text{joint},e)} = \frac{K_{l,n}^{(\text{joint},e)}}{T^{(\text{joint},e)}}$. We apply the selector directly to these pooled probabilities to obtain joint scores $s_{l,n}^{(\text{joint},e)}$, and select the top- r neurons. This approach is data-efficient and treats cross-lingual evidence as natural augmentation, but may be dominated by high-resource languages if case counts are imbalanced across $\mathcal{L}$.

Consistency-Regularized Fusion (CR-Fusion). To explicitly prefer cross-lingual stability, we penalize dispersion of scores across languages. $\tilde{s}_{l,n}^{(\ell,e)} = (s_{l,n}^{(\ell,e)} - \mu^{(\ell)}) / (\sigma^{(\ell)} + \epsilon_0)$, where $\mu^{(\ell)}$ and $\sigma^{(\ell)}$ are the mean and standard deviation computed over all (layer, neuron, emotion) triplets.

$$\mu_{l,n}^{\text{CR},(e)} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \tilde{s}_{l,n}^{(\ell,e)},$$

$$\sigma_{l,n}^{\text{CR},(e)} = \sqrt{\frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} (\tilde{s}_{l,n}^{(\ell,e)} - \mu_{l,n}^{\text{CR},(e)})^2}.$$

We then score neurons by: $s_{l,n}^{\text{CR},(e)} = \mu_{l,n}^{\text{CR},(e)} - \lambda \cdot \sigma_{l,n}^{\text{CR},(e)}$, where $\lambda \geq 0$ acts as a regularization parameter that penalizes cross-lingual variance. Higher values of λ prioritize neurons with high stability over those with high absolute monolingual scores, thereby filtering out language-specific units. Sensitivity to λ is discussed in § 6.2.

3.4 The Estimand of Cross-Lingual Fusion

Fusion Strategies as a Robustness Family. Write the per-language selector score for a fixed neuron as $s^{(\ell,e)} = \theta(e) + \delta^{(\ell,e)} + \xi^{(\ell,e)}$, where $\theta(e)$ is a language-invariant emotion effect, $\delta^{(\ell,e)}$ is genuine language-specific deviation with $\mathbb{E}_\ell[\delta^{(\ell,e)}] = 0$ and variance τ^2 , and $\xi^{(\ell,e)}$ is finite-sample noise. Under a Gaussian approximation, the CR-Fusion objective $\mu^{\text{CR}} - \lambda \sigma^{\text{CR}}$ is the $(1-q)$ lower confidence bound of the neuron’s selectivity in a randomly drawn language, with $\lambda = \Phi^{-1}(1-q)$. The three fusion strategies of §3.3 are therefore one family indexed by robustness: mean fusion ($\lambda=0$), of which Joint Fusion is the token-weighted counterpart, targets expected transfer to a new language,

CR-Fusion targets quantile transfer, and Overlap Fusion targets worst-case transfer.

The Predicted Optimal Penalty. Our evaluation averages ESS over held-out languages, which constitutes an expected-transfer objective. If languages are exchangeable, the Bayes-optimal selection criterion for this objective is θ itself, whose minimum-variance unbiased estimator is the cross-language mean; the predicted optimum is therefore λ near zero. Moreover, the observed dispersion estimates $\tau^2 + \bar{v}$, where $\bar{v}$ is within-language sampling variance: since $\hat{P}$ is a binomial proportion, $v \approx P(1-P)/T^{(\ell,e)}$, and T varies by more than an order of magnitude across our conditions (Appendix A.5). A large penalty potentially removes neurons whose statistics are estimated from sparse data rather than neurons that are meaningfully language-specific. Moreover, because selection at a fixed top- r fraction operates in the upper tail, where score magnitude and dispersion are positively correlated, a large penalty drives the selected set toward uniformly weak units. Both predictions are confirmed in §6.2.

The Recovered Estimand. Decompose the activation-probability tensor additively over languages and emotions,

$$P_{l,n}^{(\ell,e)} = m_{l,n} + u_{l,n}^{(\ell)} + b_{l,n}^{(e)} + \gamma_{l,n}^{(\ell,e)} + \varepsilon_{l,n}^{(\ell,e)}, \quad (2)$$

where m is the neuron’s baseline activity, $u^{(\ell)}$ a language main effect (overall firing shift), $b^{(e)}$ the language-invariant emotion effect, and $\gamma^{(\ell,e)}$ the language $\times$ emotion interaction. For a complete design with uniform weights, the least-squares fit satisfies $\hat{m} + \hat{b}^{(e)} = \frac{1}{|\mathcal{L}|} \sum_{\ell} P^{(\ell,e)}$, the unweighted cross-language mean profile. Joint Fusion scores the token-weighted pooled profile $\sum_{\ell} K^{(\ell,e)} / \sum_{\ell} T^{(\ell,e)}$, and small- λ CR-Fusion approximates it.

Empirically, applying ConAct to $m + \hat{b}$ selects neuron sets with mean Jaccard similarity 0.98 against Joint Fusion on the two models with complete language–emotion coverage, dropping to 0.41–0.63 on the two models with missing cells, where the weighted fit and naive pooling legitimately diverge (Appendix B). Cross-lingual fusion thus succeeds not by consensus filtering but by estimating a specific, identifiable estimand: the invariant emotion effect, which no single-corpus procedure can isolate.

Magnitude of the Invariant Share. Noise-corrected variance components of Eq. (2) yield invariant shares between 0.30 (Qwen2.5-Omni-7B) and 0.59 (MiniCPM-o-4.5) among top-ranked emotion-selective neurons (Appendix B). In no model does the invariant share exceed 0.6: much of the emotion-conditioned activation structure, in two models the majority, is specific to individual languages or corpora (see Limitations).

3.5 Intervention

Given selected neuron indices $\mathcal{I}_l^{(e)}$ for emotion e at layer l (monolingual or fused), we evaluate causal effects via two interventions: deactivation and steering. Our interventions operate on the post-activation SwiGLU gate outputs in decoder MLP modules. Let $g_{l,t} \in \mathbb{R}^{D_l}$ denote the activated gate vector at layer l and token position t , computed as $g_{l,t} = \text{act}(\text{gate_proj}(x_{l,t}))$, where $\text{act}(\cdot)$ is the activation function.

Deactivation. We zero selected neurons’ activations through element-wise masking:

$$de_{l,n}^{(e)} = \begin{cases} 0, & n \in \mathcal{I}_l^{(e)} \\ 1, & \text{otherwise} \end{cases}, \quad \tilde{g}_{l,t}^{\text{deact}} = g_{l,t} \odot de_{l,n}^{(e)}.$$

Steering. We amplify selected neurons by applying a multiplicative gain factor $\alpha \geq 0$:

$$st_{l,n}^{(e)}(\alpha) = \begin{cases} 1 + \alpha, & n \in \mathcal{I}_l^{(e)} \\ 1, & \text{otherwise} \end{cases}, \quad \tilde{g}_{l,t}^{\text{steer}} = g_{l,t} \odot st_{l,n}^{(e)}(\alpha).$$

The modified gate activations $\tilde{g}_{l,t}$ are then used in place of $g_{l,t}$. We apply these interventions systematically across evaluated languages to assess whether MLENs induce aligned causal changes, thereby validating their cross-lingual generalizability and functional specificity.

4 Experiment Setup

Datasets and Partition. Our test bed comprises 12 monolingual emotional speech datasets categorized by their role in the transferability pipeline. The **Identification Set** ($\mathcal{L}_{\text{id}}$) consists of languages used during activation logging to ensure cross-lingual consistency: Amharic (ASED (Retta et al., 2023)), Moroccan Arabic (MDER (amine Soumiaa, 2024)), Bengali (SUBESCO (Sultana et al., 2021)), English (MSP-Podcast (Busso et al., 2026)), Italian (Emozionalmente (Catania et al., 2025)), Mandarin (BIIC (Upadhyay et al., 2023)), Polish (nEMO

(Christop, 2024)), and Urdu (UrduSER (Akhtar et al., 2025)). The **Held-out Set** ($\mathcal{L}_{\text{held}}$) contains languages reserved strictly for zero-shot evaluation: French (CaFE (Gournay et al., 2018)), German (EmoDB (Burkhardt et al., 2025)), Persian (ShEMO (Mohamad Nezami et al., 2019)) and Russian (RESO (Amentes et al., 2023)). Notably, Amharic, Bengali, Urdu, and Moroccan Arabic are treated as low-resource languages within this framework (Koehn and Knowles, 2017). The dataset statistics are provided in Appendix A Table 3. Evaluation partitions sample up to 150 utterances per emotion (Appendix A.2).

Models, Prompting and Decoding. We evaluate four open-sourced LALMs: Audio-Flamingo-3 (Goel et al., 2025), Kimi-audio (KimiTeam, 2025), MiniCPM-o-4.5 (Yao et al., 2024), and Qwen2.5-Omni-7B (Xu et al., 2025b). All selected models designedly support English and Mandarin, with varying, officially unclaimed capabilities in other languages. All models are evaluated in a controlled multiple-choice format using a single instruction template (Appendix A.3). All inference uses deterministic decoding (greedy, temperature=0); reported variance therefore reflects cross-language variability (Appendix A.4).

Metrics. We assess baseline performances using unweighted average recall (UAR) across five emotions: anger, fear, happiness, neutral, and sadness. To quantify causal effects, we define **Self-Effects (SE)** as the accuracy change on a specific emotion subset when applying the matched neuron mask, relative to the unintervened baseline: $\text{SE}(e) = \text{Acc}_{e \rightarrow e} - \text{Acc}_{\text{unintervened}}(e)$. **Average Cross-Effects (ACE)** measures, for each target emotion e , the mean accuracy change on e under masks matched to other emotions: $\text{ACE}(e) = \frac{1}{|\mathcal{E}| - 1} \sum_{e' \neq e} (\text{Acc}_{e' \rightarrow e} - \text{Acc}_{\text{unintervened}}(e))$. **Emotion Selectivity Score (ESS)** quantifies specificity by comparing self-effect against cross-effect. The per-emotion ESS is defined as: $\text{ESS}(e) = \text{SE}(e) - \text{ACE}(e)$. The global ESS, which aggregates across all valid emotions, is computed as: $\text{ESS} = \frac{1}{|\mathcal{E}_{\text{valid}}|} \sum_{e \in \mathcal{E}_{\text{valid}}} \text{ESS}(e)$, where $\mathcal{E}_{\text{valid}}$ denotes emotions with at least one test instance. Unless otherwise stated, all reported ESS values refer to this global metric. Negative ESS under deactivation indicates selective impairment of target emotions while preserving recognition of other emotions; positive ESS under steering indicates selective enhancement of target emotions.

5 Results

5.1 Cross-Lingual Agreement of Monolingual Emotion-Sensitive Neurons

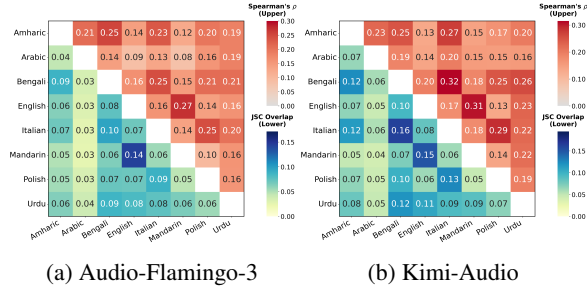


Figure 1: Cross-lingual agreement of monolingually-identified ESNs across two models (see other two in Appendix D.1). Each heatmap displays JSC (lower triangle), measuring neuron set intersection, and Spearman's ρ (upper triangle), measuring agreement in selectivity score rankings. All neurons selected using ConAct with $r=0.5\%$.

We first examine whether ESNs identified independently per language exhibit cross-lingual agreement in both ranking structure and set membership. Figure 1 present pairwise Jaccard Similarity Coefficient (JSC) and Spearman's rank correlation (ρ) for all four LALMs across eight identification languages. Across all models, we observe a consistent dissociation: the exact set intersection of top-ranked neurons (discrete identity overlap) is minimal (JSC predominantly below 0.10, with occasional pairs reaching 0.15–0.16), while the underlying selectivity scores retain a shared ranking structure (ρ mostly between 0.10 and 0.25 for Audio-Flamingo-3 and Kimi-Audio, and up to ≈ 0.6 for MiniCPM-o-4.5).

5.2 Saturation of Monolingual Evidence

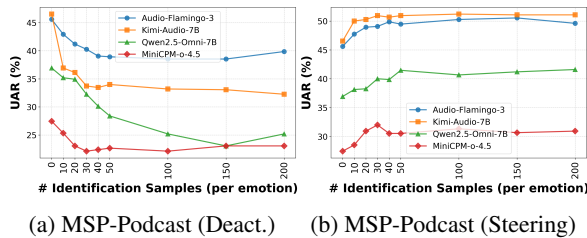


Figure 2: Saturation of monolingual evidence: UAR (%) under intervention as identification sample size grows (MSP-Podcast; BIIC in Appendix D.2 shows the same plateau beyond ~ 50 instances).

We evaluate the causal properties of identified neurons through functional intervention. To establish monolingual baselines before introducing fusion, Figure 2 shows the causal efficacy of ESN

masks identified using varying numbers of correctly predicted instances from the Identification Set ($\mathcal{L}_{id}$). Across all evaluated models, intervention effects plateau rapidly beyond approximately 50 instances.

The bottleneck for isolating transferable affective units is therefore the linguistic scope of the identification data rather than sample volume, making cross-lingual evidence integration essential. Accordingly, all subsequent experiments use a fixed budget of $c=50$ instances per emotion to focus on the causal benefits of cross-lingual consistency (see Appendix A.5 for exact counts and justification).

5.3 Causal Validation of MLENs Through Targeted Intervention

We evaluate the functional necessity and influence of the identified neurons through two targeted interventions: deactivation (to assess necessity for recognition) and activation steering (to assess the neurons' capacity to modulate model behavior).

Deactivation. To assess whether identified neurons contribute causally to emotion recognition, we measure the reduction in ESS under deactivation, which are summarized in Table 1 across two language groups. CR-Fusion ($\lambda=0.3$) exceeds the best monolingual mask for three models in both groups. Joint Fusion does so on Audio-Flamingo-3 and Kimi-Audio, but falls short of the strongest monolingual mask on MiniCPM-o-4.5 (-6.26 vs. -7.64) and Qwen2.5-Omni-7B (-6.05 vs. -6.99), consistent with its token-weighted pooling being dominated by high-resource conditions. Overlap Fusion is the weakest fusion strategy on three of four models and reverses sign on Qwen2.5-Omni-7B ($+0.58$); Audio-Flamingo-3 is the exception, where the consensus set remains strongly causal.

Steering. Table 2 shows steering effects across two language groups. CR-Fusion ($\lambda=0.3$) yields superior steering efficacy in $\mathcal{L}_{held}$ for three of the four LALMs. For MiniCPM-o-4.5, CR-Fusion achieves $+6.48$ pp in held-out languages, substantially outperforming English ($+4.70$ pp) and Mandarin ($+5.48$ pp) baselines. The exception is Qwen2.5-Omni-7B, where the Mandarin mask remains marginally stronger ($+3.97$ vs. $+3.64$ pp); this is consistent with Qwen having the lowest language-invariant share (§3.4, Appendix B), so there is less invariant structure for fusion to recover. Together, these results provide causal evidence that CR-Fusion identifies functionally important, language-shared affective units.

LALM	Group	Monolingual ESN Masks								Fusion MLEN Masks		
		English	Mandarin	Amharic	Arabic	Bengali	Italian	Polish	Urdu	Overlap	Joint	CR ($\lambda=0.3$)
Audio Flamingo 3	$\mathcal{L}_{id}$	-5.92	-3.93	-3.85	-1.43	-2.92	-2.03	-4.70	-6.16	-10.15	-9.20	-9.92
	$\mathcal{L}_{held}$	-5.64	-4.84	-3.31	-1.34	-3.44	-1.64	-4.33	-4.62	-8.86	-9.57	-9.93
	Avg.	-5.83 (2.05)	-4.24 (1.26)	-3.67 (1.16)	-1.40 (1.06)	-3.09 (0.95)	-1.90 (0.65)	-4.58 (1.13)	-5.65 (2.20)	-9.72 (4.17)	-9.33 (2.30)	-9.92 (2.57)
Kimi -Audio	$\mathcal{L}_{id}$	-13.00	-12.05	-7.22	-3.83	-10.81	-12.80	-4.85	-12.97	-6.93	-16.68	-16.75
	$\mathcal{L}_{held}$	-12.80	-12.54	-5.82	-3.63	-9.98	-13.64	-3.47	-15.02	-7.66	-17.58	-19.31
	Avg.	-12.93 (3.68)	-12.21 (3.54)	-6.75 (2.40)	-3.76 (1.42)	-10.53 (2.85)	-13.08 (3.67)	-4.39 (1.77)	-13.66 (3.13)	-7.18 (2.03)	-16.98 (4.28)	-17.60 (4.50)
MiniCPM -o-4.5	$\mathcal{L}_{id}$	-6.02	-6.72	-2.07	-0.57	-2.88	-2.61	-2.68	-6.22	-4.39	-5.62	-7.29
	$\mathcal{L}_{held}$	-7.86	-9.47	-3.41	-0.74	-3.36	-4.41	-2.83	-8.29	-5.35	-7.56	-9.51
	Avg.	-6.63 (2.98)	-7.64 (3.99)	-2.52 (2.09)	-0.63 (0.55)	-3.04 (1.34)	-3.21 (2.15)	-2.73 (2.40)	-6.91 (3.89)	-4.71 (2.34)	-6.26 (3.49)	-8.03 (4.42)
Qwen2.5 -Omni-7B	$\mathcal{L}_{id}$	-7.09	-6.28	-6.70	-0.83	-3.10	-2.37	-0.28	-1.97	0.68	-5.91	-6.68
	$\mathcal{L}_{held}$	-6.77	-5.91	-6.02	-0.57	-3.38	-3.82	0.28	-3.84	0.40	-6.34	-7.86
	Avg.	-6.99 (2.28)	-6.16 (1.82)	-6.48 (1.17)	-0.74 (1.54)	-3.19 (1.95)	-2.85 (1.31)	-0.09 (1.16)	-2.59 (1.90)	0.58 (1.64)	-6.05 (0.78)	-7.07 (2.16)

Table 1: Deactivation effects on ESS. “Avg.” rows denote means across all languages regardless of group; standard deviation in parentheses. Negative values indicate reduced specificity, and color intensity indicates effect magnitude. Per-language results in Appendix D.5 Table 8.

LALM	Group	Monolingual ESN Masks								Fusion MLEN Masks		
		English	Mandarin	Amharic	Arabic	Bengali	Italian	Polish	Urdu	Overlap	Joint	CR ($\lambda=0.3$)
Audio Flamingo 3	$\mathcal{L}_{id}$	3.72	2.11	1.89	0.45	1.51	0.91	2.82	3.60	2.22	4.50	4.68
	$\mathcal{L}_{held}$	2.72	2.04	1.66	0.28	1.40	1.19	1.73	2.25	1.51	3.80	4.00
	Avg.	3.39 (1.11)	2.09 (0.63)	1.81 (0.90)	0.39 (0.41)	1.48 (0.59)	1.00 (0.54)	2.46 (1.08)	3.15 (1.20)	1.98 (0.88)	4.27 (1.21)	4.45 (1.26)
Kimi -Audio	$\mathcal{L}_{id}$	5.36	5.19	3.22	2.12	4.80	5.08	2.19	5.58	3.28	7.25	7.25
	$\mathcal{L}_{held}$	4.70	4.53	2.43	1.96	4.15	4.05	1.70	5.15	3.39	6.29	6.34
	Avg.	5.14 (1.58)	4.97 (1.57)	2.96 (1.30)	2.07 (1.05)	4.58 (1.72)	4.74 (1.46)	2.03 (1.09)	5.44 (1.45)	3.32 (1.04)	6.93 (2.02)	6.95 (1.74)
MiniCPM -o-4.5	$\mathcal{L}_{id}$	4.18	5.15	1.98	0.26	2.10	2.21	2.81	5.17	2.59	4.29	5.97
	$\mathcal{L}_{held}$	4.70	5.48	2.65	-0.09	1.82	2.66	1.92	5.24	2.74	3.88	6.48
	Avg.	4.35 (1.58)	5.26 (1.88)	2.20 (1.39)	0.15 (0.43)	2.01 (0.67)	2.36 (1.15)	2.51 (1.26)	5.20 (2.07)	2.64 (1.00)	4.15 (1.24)	6.14 (2.48)
Qwen2.5 -Omni-7B	$\mathcal{L}_{id}$	3.94	3.68	3.35	0.26	1.28	1.13	0.24	1.45	-0.22	3.87	3.39
	$\mathcal{L}_{held}$	3.14	3.97	3.01	0.15	0.23	1.91	-0.01	2.03	0.29	3.58	3.64
	Avg.	3.68 (0.93)	3.78 (0.81)	3.24 (0.77)	0.23 (0.69)	0.93 (1.35)	1.39 (0.60)	0.16 (0.94)	1.65 (0.96)	-0.05 (1.01)	3.77 (0.95)	3.47 (1.14)

Table 2: Steering effects on ESS with $\alpha=0.5$. “Avg.” rows denote means across all languages regardless of group; standard deviation in parentheses. Positive values indicate enhanced specificity, and color intensity indicates effect magnitude. Per-language results in Appendix D.5 Table 9.

6 Discussion

6.1 Emotion-Level Heterogeneity in Cross-Lingual Generalization

Causal interventions reveal substantial heterogeneity across emotions. Anger, happiness, and sadness exhibit the largest ESS magnitudes across all evaluated LALMs under both deactivation and steering, and this ordering is preserved across the identification and held-out groups (Appendix A.6). Fear and neutral show weaker and more variable effects; for fear, this plausibly reflects acoustic overlap with high-arousal anger and a floor effect from near-zero baseline accuracy in several model–language pairs (Appendix D.4), which leaves few correctly predicted instances available for identification (Appendix A.5) rather than a shortage of fear utterances in the corpora themselves (Table 6). Strong causal potency does not, however, imply more language-invariant encoding: per-emotion invariant shares

are highest for neutral and fear, so the causal ordering is better explained by baseline accuracy and class support than by representational universality.

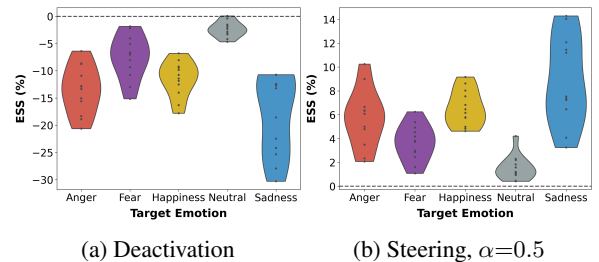


Figure 3: ESS distributions under (a) deactivation and (b) steering, comparing CR-Fusion ($\lambda=0.3$) against monolingual baselines across all evaluated emotions. Per-emotion magnitudes are provided in Appendix A.6.

Independently of these differences, Figure 3 shows that CR-Fusion ($\lambda=0.3$) matches or exceeds monolingual baselines for every emotion in both groups. Consistent with §3.4, this advantage does not come from suppressing cross-lingual variance

(the penalty is near zero at the operating point) but from pooling multiple corpora to estimate the language-invariant component.

6.2 The Impact of Consistency Penalty Strength

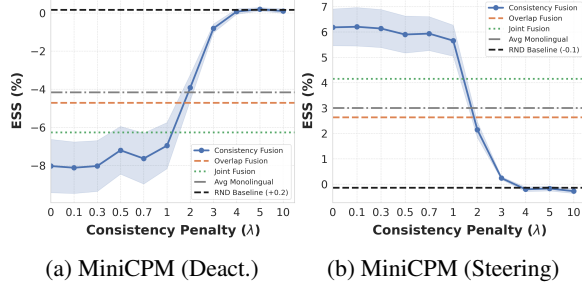


Figure 4: Sensitivity of intervention effects to consistency penalty λ . Average ESS (± 1 standard error of the mean across 12 languages, shown as the shaded band) under (a) deactivation and (b) steering for MiniCPM. Results for other LALMs are provided in Appendix D.

We analyze the sensitivity of causal effects to the consistency penalty λ , which trades aggregate selector strength (μ) against cross-lingual stability (low σ). Figure 4 and Appendix D.3 Figure 9 show that intervention effects are strongest at small λ and degrade sharply beyond $\lambda \approx 1$, approaching the random-selection baseline by $\lambda \approx 3$ for all four LALMs; for Qwen2.5-Omni-7B under deactivation the effect reverses sign, indicating that at large λ the selected units are no longer emotion-selective at all. We therefore adopt $\lambda=0.3$ as the operating point, where the penalty is active but weak.

This profile favors both predictions of §3.4: the optimal penalty is near zero under the expected-transfer objective, and large λ preferentially discards low-resource conditions while shifting selection toward uniformly weak units. Overlap Fusion is correspondingly the weakest fusion strategy in Table 2 for all four models and in Table 1 for three.

6.3 Language Contributions and Asymmetric Transfer

We use leave-one-out to test whether the multilingual mask is driven by a few dominant identification languages: for each identification language, we remove it from the fusion pool, re-identify the CR-Fusion mask, and measure the change in ESS on each evaluation language (Figure 5). Under deactivation, no single removal explains the gains of CR-Fusion; removals instead produce consistent but non-uniform ESS reductions across evaluation languages, including when low-resource languages

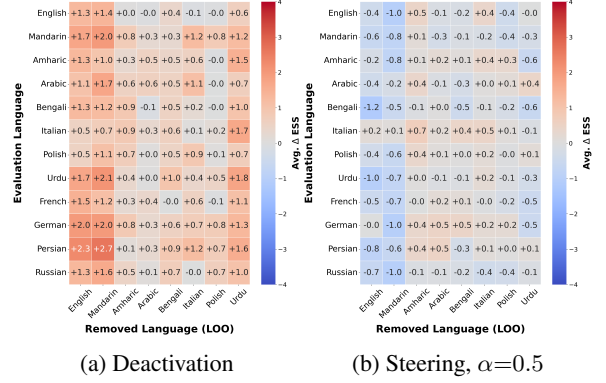


Figure 5: Leave-one-out Δ ESS heatmaps averaged over four models: effect of excluding each identification language (columns) on each evaluation language (rows).

are removed. The fused mask thus aggregates partially complementary evidence, with low-resource corpora supplying non-redundant signal for selecting transferable units. Steering shows the same qualitative pattern with smaller and more localized changes. Finally, because each language is represented by a single corpus, language identity and recording condition (elicitation style, channel, speakers) are confounded; we therefore read it as evidence of non-redundant cross-corpus contributions rather than a typological transfer hierarchy.

7 Conclusion

This work presents the first systematic mechanistic investigation of multilingual emotion representations in LALMs. Through causal interventions across 12 typologically diverse languages and four modern LALMs, we showed that monolingually identified emotion neurons share little set-level overlap and that monolingual evidence saturates rapidly, motivating multilingual identification; that CR-Fusion recovers a cross-lingually shared emotion component inaccessible to any single-corpus procedure, yielding more transferable causal control in zero-shot and low-resource settings; and that leave-one-out ablation reveals asymmetric, non-substitutable language contributions, with low-resource corpora both supplying non-redundant identification evidence and being among those that benefit most from fusion.

By demonstrating shared affective representations that generalize across diverse spoken languages, this work provides a mechanistic foundation for equitable affective computing, enabling speech emotion recognition systems that can serve low-resource language communities without requiring extensive language-specific training data.

Limitations

Our language stratification reflects functional roles in the experimental pipeline rather than strict claims about model exposure, as LALMs may have encountered these languages implicitly during web-crawled pre-training. Additionally, our analysis addresses discrete emotion categories; extending to dimensional affect models or fine-grained emotional states remains for future work.

We also admit that each language in our study is represented by a single corpus, so language identity and recording condition (elicitation style, channel, speaker population) are confounded by construction. A post-hoc analysis of the language $\times$ emotion interaction component of Eq. (2) indicates that similarity in language-conditioned activation structure tracks corpus elicitation type (acted vs. naturalistic) at least as strongly as typological relatedness; the asymmetric transfer patterns of §6.3 should therefore be read as corpus-level rather than purely linguistic effects. Disentangling the two requires at least one language represented by corpora of different elicitation types, which we leave to future work. Third, all fusion strategies studied here recover only the language-invariant component of the emotion code; the language-specific component, which accounts for 41–70% of emotion-conditioned activation variance across models, is discarded, and exploiting it for target-aware adaptation remains open.

Ethical Considerations and Broader Societal Impact

While our work aims to advance mechanistic understanding of emotion processing in multimodal foundation models, we acknowledge potential risks associated with neuron-level emotion manipulation techniques. The intervention methods demonstrated in this study could be misused to artificially induce or suppress emotional signals in generated speech, enabling more persuasive synthetic media or deceptive voice-based systems. Furthermore, fine-grained understanding of ESNs could facilitate targeted attacks on emotion recognition systems or be exploited to bypass affective content moderation. We emphasize that our interventions are evaluated only in controlled research settings on pre-trained models and are not packaged or evaluated as a deployment system; nevertheless, the underlying techniques could be employed in undesirable ways. We hence encourage the research

community to develop appropriate safeguards, such as detection mechanisms for artificially manipulated emotional content, alongside continued interpretability research.

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A Reproducibility

A.1 Computational Resources

All experiments were conducted using NVIDIA A100 and H100 GPUs from a shared cluster budget with 8 GPUs of each type. The total GPU consumption, including failed runs and debugging, was approximately 312 A100-hours and 687 H100-hours. The majority of compute time was dedicated to intervention experiments across 12 languages, four LALMs and multiple parameter settings, with activation logging requiring comparatively fewer resources. Each LALM required between 40-80 GB of GPU memory depending on model size and batch configuration. We estimate that reproducing only the final reported experiments would require approximately 400 A100-hours or equivalent.

A.2 Datasets and Models

For each dataset, we sample up to 150 utterances per emotion for the evaluation partition using a fixed random seed to ensure reproducibility; when a dataset contains fewer than 150 utterances for an emotion (e.g., CaFE, EmoDB, ShEMO; Table 3), all available utterances are used.

Models	Hugging Face Identifier	License
Audio-Flamingo-3	https://huggingface.co/nvidia/audio-flamingo-3	NVIDIA OneWay Noncommercial License
Kimi-Audio	https://huggingface.co/moonshotai/Kimi-Audio-7B-Instruct	Apache License 2.0, MIT License
MiniCPM-o-4.5	https://huggingface.co/openbmb/MiniCPM-o-4.5	Apache License 2.0
Qwen2.5-Omni-7B	https://huggingface.co/Qwen/Qwen2.5-Omni-7B	Apache License 2.0

Table 4: Sources and licenses for the four evaluated LALMs.

A.3 SER Prompt Template

To reduce intrinsic positional and label preference biases within LALMs, we randomize the index $\leftrightarrow$ emotion mapping per instance: for each speech clip instance, each emotion is randomly assigned to an option letter (e.g., “A”). We implement the following prompt template for SER.

Based on the provided speech clip, identify the emotion expressed in the speech. Choose the option that best matches the perceived emotion from the audio:

- A: emotion 1
- B: emotion 2
- C: emotion 3
- D: emotion 4
- E: emotion 5

Output exactly one option letter and no other text.

A.4 Decoding and Statistical Reporting

We decode deterministically (greedy; temperature 0) with a 20-token generation limit and apply lightweight post-processing to extract the option letter from model outputs. Deterministic decoding ensures that model outputs are as reproducible as possible given fixed inputs and model weights. Consequently, repeated runs on identical data produce identical results, and variance in reported metrics reflects cross-language or cross-condition variability rather than stochastic sampling noise. In Tables 1 and 2, we report mean and standard deviation (in parentheses) computed across evaluation languages within each group, quantifying the consistency of intervention effects across typologically diverse linguistic contexts. For sensitivity analyses (Figure 4), shaded bands represent ± 1 SEM computed over all 12 evaluation languages ($n=12$). Data partitioning uses fixed random seeds (specified in our released code) to ensure exact reproducibility of all reported results.

We do not report seed-based confidence intervals, as deterministic decoding renders repeated runs identical and seed variance is exactly zero. Bootstrap resampling over test utterances would quantify sampling variability of the evaluation sets;

Group	$\mathcal{L}_{id}$								$\mathcal{L}_{held}$			
Language	Amharic	Arabic	Bengali	English	Italian	Mandarin	Polish	Urdu	French	German	Persian	Russian
Datasets	ASED	MDER	SUBESCO	MSP-Podcast	Emozio-nalmente	BIIC	nEMO	UrduSER	CaFE	EmoDB	ShEMO	RESO
Angry	486	400	1000	36260	986	6543	749	500	144	137	1059	219
Fear	510	400	1000	1943	986	898	736	500	144	122	38	223
Happy	486	400	1000	58684	986	18859	749	500	144	115	201	218
Neutral	522	400	1000	79117	986	21269	809	500	72	104	1028	191
Sad	470	400	1000	24629	986	5679	769	500	144	121	449	162

Table 3: Dataset statistics showing utterance counts per emotion across 12 multilingual speech emotion languages/-datasets, grouped by experimental role: Identification and Held.

we instead report cross-language dispersion, which is the variability most relevant to our claims about multilingual generalization.

All standard deviations and standard errors are computed using closed-form formulas implemented via NumPy¹ library functions.

A.5 Identification Instance Budget and Data Availability Constraints

Our neuron identification procedure targets $c=50$ correctly predicted instances per emotion per language. However, certain model-language-emotion triples exhibit low baseline accuracy, yielding fewer correct predictions available for activation logging. In such cases, we use all available correctly predicted instances rather than discarding the emotion or language entirely.

We retain low-count conditions for two reasons. First, excluding underperforming conditions would systematically bias our analysis toward high-resource, well-recognized settings, undermining our goal of investigating multilingual emotion representations. Second, our saturation analysis (§5.2) demonstrates that intervention effects plateau beyond approximately 50 instances, indicating that reduced counts can still provide informative activation statistics.

For severely underrepresented cases such as MiniCPM-o-4.5 Fear (36 total instances, with zero for Arabic and Polish), we acknowledge reduced reliability. However, this primarily reflects MiniCPM-o-4.5’s strong bias toward Neutral predictions (98.83% baseline accuracy). The emotion-level analysis (§6.1) independently confirms that Fear exhibits weaker causal effects, consistent with data sparsity. Our primary claims rest on emotions with robust identification data across all models, namely anger, happiness, and sadness.

¹<https://numpy.org/>

A.6 Per-Emotion Detail

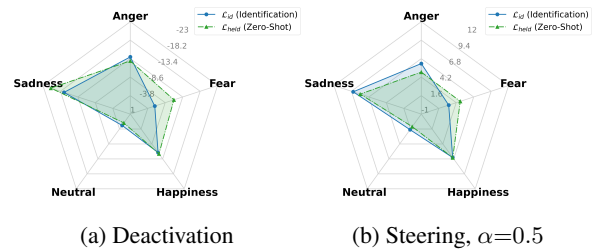


Figure 6: Per-emotion ESS magnitudes under (a) deactivation and (b) steering, comparing CR-Fusion ($\lambda=0.3$) against monolingual baselines, reported separately for $\mathcal{L}_{id}$ and $\mathcal{L}_{held}$. Radial axes are not shared across panels.

Group consistency. Figure 6 reports per-emotion ESS magnitudes separately for $\mathcal{L}_{id}$ and $\mathcal{L}_{held}$. The relative ordering of emotions is preserved across groups for all models, under both deactivation and steering: the emotions with the strongest effects on identification languages are also the strongest on languages never seen during identification. Radial axes are not shared across panels.

Causal potency versus representational invariance. It is tempting to interpret the ordering in §6.1 as evidence that anger, happiness, and sadness are encoded in more language-invariant form than fear or neutral. However, the variance decomposition of §3.4 does not support that interpretation: per-emotion invariant shares (Table 6) are not highest for the three emotions with the strongest causal effects in any model, and averaged across models the ordering is inverted: neutral (0.52) and fear (0.50) exceed anger (0.43), sadness (0.43), and happiness (0.41).

The two quantities measure different things, and the dissociation is informative. The invariant share is a property of the representation: how much of a neuron’s emotion tuning is shared across languages. ESS is a property of model behavior under inter-

vention, bounded by baseline recognition accuracy, per-emotion test support, and confusability within the response set. Fear is the least or second-least accurately recognized emotion in every model, and neutral, although nominally the most accurate in three of four models, is inflated by a strong neutral response bias (Appendix D.4): its high accuracy reflects a default prediction rather than discriminative use of emotion-sensitive units, so suppressing neutral-selective units cannot remove it. Anger and sadness are recognized well above the fear floor in every model. Baseline accuracy, response bias and per-emotion test support therefore suffice to produce the observed causal ordering without any difference in representational invariance.

B Variance Decomposition of Multilingual Emotion Coding

Estimation. We fit Eq. (2) per neuron by alternating weighted centering over the language and emotion axes of the $|L| \times |E|$ activation-probability table, after dividing each language’s tensor by its global standard deviation to remove differences in dynamic range. For complete tables the fit converges in one iteration and coincides with the two-way ANOVA solution; for tables with missing cells (emotions for which a model produced too few correct predictions in a given language, Appendix A.5) it converges to the weighted least-squares additive fit. All quantities below are computed over the top- $r=0.5\%$ of neurons ranked by the ConAct margin of the pooled profile, i.e. the population from which masks are drawn.

Noise correction. Since $\hat{P}$ is a binomial proportion, each cell carries sampling variance $v = P(1 - P)/T^{(\ell,e)}$, which inflates the apparent interaction variance far more than the invariant variance: noise enters the residual $\hat{\gamma}$ with weight $(|L|-1)(|E|-1)/(|L||E|)$ but enters $\hat{b}$ with weight only $(|E|-1)/(|L||E|)$, a factor $|L|-1$ smaller, because averaging over languages suppresses it. We therefore report shares based on $\hat{\sigma}_\gamma^2 = \text{ms}(\hat{\gamma})/f_\gamma - \bar{v}$ and $\hat{\sigma}_b^2 = (\text{ms}(\hat{b}) - \bar{v}f_b/|L|)/f_b$, where $\text{ms}(\cdot)$ is the mean square, $f_\gamma = (|L|-1)(|E|-1)/(|L||E|)$ and $f_b = (|E|-1)/|E|$ are residual projection factors, and $\bar{v}$ is the mean estimated binomial variance. Uncorrected shares are within 0.03 of corrected values for every model, and estimated noise accounts for only 13–22% of the raw interaction energy, so the language-specific component is not a sampling artifact.

LALM	$\hat{\sigma}_b^2$ ($\times 10^3$)	$\hat{\sigma}_\gamma^2$ ($\times 10^3$)	Share (raw)	Share (corr.)	JSC
Audio-Flamingo-3	3.57	4.70	0.44	0.43	0.98
Kimi-Audio	2.47	2.27	0.50	0.52	0.98
MiniCPM-o-4.5	2.10	1.45	0.59	0.59	0.41
Qwen2.5-Omni-7B	1.14	2.61	0.31	0.30	0.63

Table 5: Variance components of Eq. (2) among top-ranked emotion-selective neurons ($r=0.5\%$, ConAct, $c=50$). “Share” is the invariant fraction $\sigma_b^2/(\sigma_b^2 + \sigma_\gamma^2)$, raw and noise-corrected. “JSC” is the mean Jaccard similarity between masks selected on the fitted invariant component $m + \hat{b}$ and Joint Fusion masks: near-identity (0.98) for the two models with complete language–emotion tables, as predicted by the algebraic equivalence of §3.4, with divergence appearing only where missing cells make the weighted fit differ from naive pooling.

LALM	Ang.	Fear	Hap.	Neu.	Sad.
Audio-Flamingo-3	0.39	0.42	0.42	0.50	0.42
Kimi-Audio	0.54	0.50	0.44	0.61	0.48
MiniCPM-o-4.5	0.57	0.72	0.49	0.60	0.52
Qwen2.5-Omni-7B	0.23	0.35	0.30	0.34	0.29
Mean	0.43	0.50	0.41	0.52	0.43

Table 6: Per-emotion noise-corrected invariant share. The share is *not* highest for anger, happiness, and sadness in any model; averaged across models, neutral and fear show the highest invariant shares. Representational universality therefore does not explain the causal-potency ordering of §6.1 (Appendix A.6).

Interpretation. From Table 5 and 6, we observe that: (1) the invariant share is below 0.6 across models; in two of four it is below one half. The language-invariant emotion code that fusion isolates is real and causally potent (§5), but it is a minority-to-half share of the emotion-conditioned activation structure; (2) The masks selected on the fitted invariant component coincide with Joint Fusion masks wherever the design is complete (JSC 0.98), confirming empirically that Joint Fusion and small- λ CR-Fusion estimate the invariant component rather than performing consensus filtering; (3) With only four models we do not treat the invariant share as a predictor of fusion gain, and the two orderings are not monotonically related: MiniCPM-o-4.5 has the highest invariant share (0.59) but one of the smallest deactivation margins over the best monolingual mask (0.39 pp), whereas Kimi-Audio and Audio-Flamingo-3 have lower shares and larger margins. The one systematic observation we draw is that Qwen2.5-Omni-7B, with the

lowest share (0.30), is the only model for which a monolingual mask (Mandarin) matches fusion under steering. We report this as a consistency check on the decomposition, not as evidence of a dose–response relation between invariant share and fusion benefit.

C Extended Related Work

Controllable Affective Speech Synthesis. A complementary line of research controls paralinguistic affect in speech generation through learned style representations (Skerry-Ryan et al., 2018; Wang et al., 2018; Wu et al., 2019; Liu et al., 2021) and continuous control variables (Lorenzo-Trueba et al., 2018; Lei et al., 2021; Xie et al., 2025). These methods control affect via architectural conditioning learned during training, whereas we intervene training-free on neurons of pre-trained LALMs.

Neuron Attribution in Multimodal Models. Beyond the audio modality, neuron-level dissection in multimodal foundation models has focused on modality-specific functional attributions (Huang et al., 2024b; Fang et al., 2024; Xu et al., 2025a; Huo et al., 2024), and audio interpretability studies rely on layer-wise probing of phonetic, speaker, and prosodic cues (Singla et al., 2022; Ulgen et al., 2024; Akman et al., 2025).

D Supplementary Results

D.1 Cross-Lingual Agreement of Monolingual Emotion-Sensitive Neurons

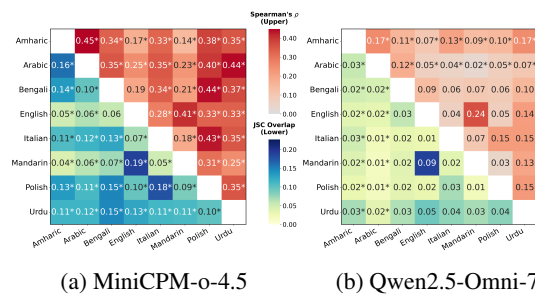


Figure 7: Cross-lingual agreement of monolingually-identified ESNs for MiniCPM-o-4.5 and Qwen2.5-Omni-7B.

D.2 Saturation of Monolingual Evidence (BIIC)

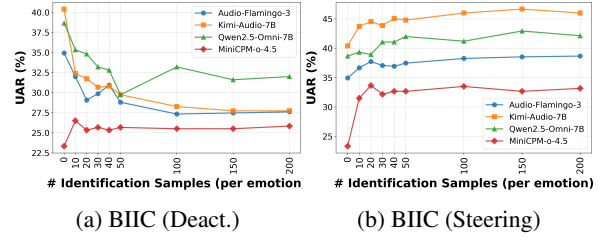


Figure 8: Saturation of monolingual evidence: UAR (%) under intervention as identification sample size grows (BIIC).

D.3 Sensitivity Analysis for Consistency Penalty

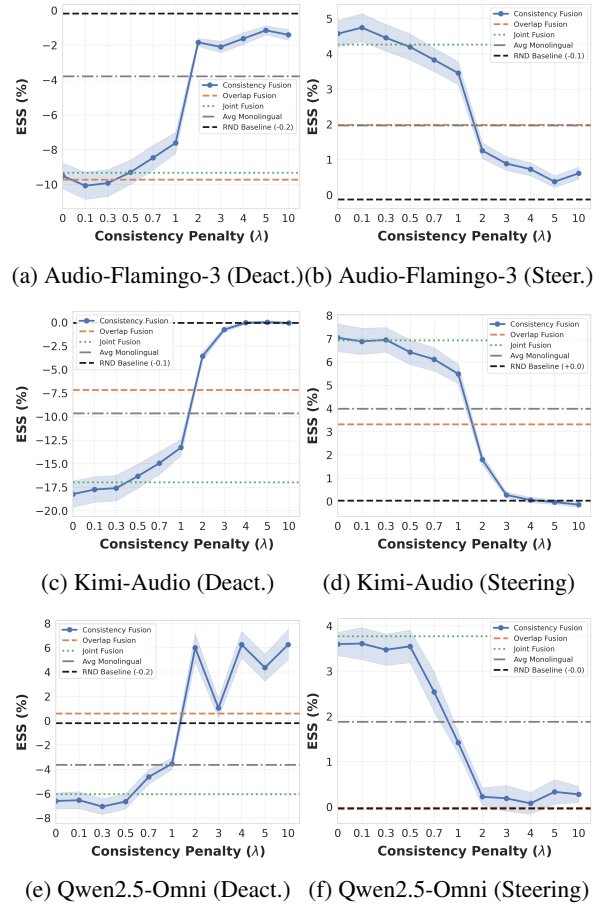


Figure 9: Sensitivity to consistency penalty λ under deactivation (left) and steering ($\alpha=0.5$, right), complementing Figure 4: (a,b) Audio-Flamingo-3, (c,d) Kimi-Audio, (e,f) Qwen2.5-Omni-7B. Shaded bands show ± 1 SEM across 12 languages.

D.4 Unintervened Baseline Results

Table 7 shows the baseline SER accuracies for unintervened models. Test set class distributions vary

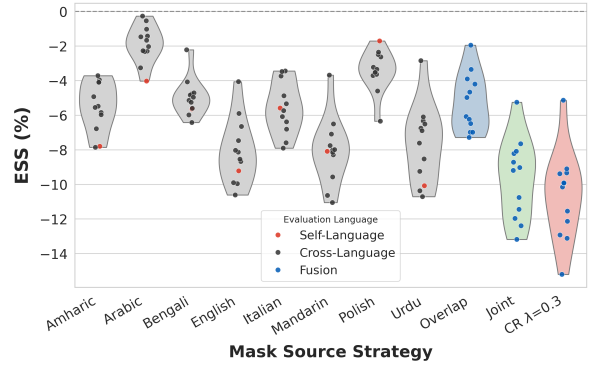
LALM	Emotion	English	Mandarin	Amharic	Arabic	Bengali	Italian	Polish	Urdu	French	German	Persian	Russian	Avg.
Audio Flamingo 3	Anger	70.00	57.33	82.00	36.67	90.00	35.33	10.67	86.00	93.75	100.00	95.33	75.33	69.37
	Fear	11.33	16.67	24.00	8.67	16.67	22.67	3.33	3.33	47.92	57.38	44.74	4.00	21.72
	Happiness	35.33	7.33	11.33	5.33	14.00	16.00	5.33	11.33	11.11	70.43	58.67	8.67	21.24
	Neutral	82.00	78.67	68.67	89.33	78.67	98.00	96.67	54.00	79.17	99.04	84.00	76.67	82.07
	Sadness	29.33	14.67	43.33	13.33	41.33	27.33	8.00	35.33	56.94	90.08	69.33	22.67	37.64
	UAR	45.60	34.93	45.87	30.67	48.13	39.87	24.80	38.00	57.78	83.39	70.41	37.47	46.41
Kimi-Audio	Anger	60.00	38.67	78.00	50.00	87.33	57.33	11.33	91.33	90.28	94.16	82.67	66.67	67.31
	Fear	8.67	14.00	19.33	16.67	40.00	33.33	3.33	6.00	53.47	35.25	55.26	11.33	24.72
	Happiness	49.33	35.33	53.33	46.67	65.33	66.00	28.00	29.33	31.94	48.70	62.67	40.00	46.39
	Neutral	80.67	84.00	94.00	58.00	68.00	90.67	100.00	76.67	83.33	99.04	74.67	83.33	82.70
	Sadness	34.00	30.00	70.67	59.33	76.67	62.00	30.67	70.67	78.47	80.17	75.33	46.00	59.50
	UAR	46.53	40.40	63.07	46.13	67.47	61.87	34.67	54.80	67.50	71.46	70.12	49.47	56.12
MiniCPM -o-4.5	Anger	14.00	5.33	10.67	5.33	58.00	8.67	0.67	34.00	38.19	21.17	22.67	25.33	20.34
	Fear	1.33	1.33	1.33	0.00	1.33	0.00	0.00	0.00	2.78	0.00	0.00	1.33	0.79
	Happiness	20.67	8.67	52.67	4.67	8.00	28.00	5.33	8.00	25.69	15.65	20.67	14.00	17.67
	Neutral	98.67	99.33	100.00	98.67	93.33	100.00	99.33	99.33	100.00	100.00	99.33	98.00	98.83
	Sadness	2.67	2.00	22.67	0.00	36.00	5.33	0.00	23.33	30.56	2.48	30.67	8.67	13.70
	UAR	27.47	23.33	37.47	21.73	39.33	28.40	21.07	32.93	39.44	27.86	34.67	29.47	30.26
Qwen2.5- Omni-7B	Anger	54.00	52.67	50.67	48.67	46.67	8.67	32.67	64.00	40.97	32.85	55.33	44.00	44.26
	Fear	13.33	24.67	2.00	2.67	2.67	2.00	2.00	5.33	4.86	10.66	7.89	8.00	7.17
	Happiness	52.67	46.00	17.33	24.00	26.00	24.67	21.33	34.67	17.36	26.09	52.67	42.67	32.12
	Neutral	24.67	38.00	19.33	15.33	18.00	28.00	16.00	12.00	22.22	21.15	10.00	28.67	21.11
	Sadness	40.00	32.00	47.33	36.67	44.00	44.67	40.00	48.67	35.42	39.67	53.33	36.67	41.53
	UAR	36.93	38.67	27.33	25.47	27.47	21.60	22.40	32.93	24.17	26.08	35.85	32.00	29.24

Table 7: Baseline SER accuracy (%) across 12 languages with deterministic decoding. “UAR” rows report unweighted average recall computed over available emotions; “Avg.” column reports UAR across languages.

across datasets, with some emotions (e.g., fear in Persian) having few or zero instances. Global ESS is computed only over emotions with available test cases; per-emotion breakdowns in Appendix D.5 assess robustness across class frequencies. We note that deactivation effects are bounded by baseline accuracy, creating potential floor effects for low-accuracy emotions. Fear’s weak deactivation effect in MiniCPM-o-4.5 (baseline $\approx 1\%$) should be interpreted cautiously given this constraint.

D.5 Per-Language Detailed Results

Table 8 and 9 provide per-language intervention results. Figure 10 displays ESS distributions across mask sources and evaluation settings under deactivation, averaged over four LALMs. Monolingual masks show substantial variability, with high-resource anchors (English, Mandarin) and certain low-resource languages (Bengali, Urdu) achieving stronger effects than others (Amharic, Arabic). Consistency fusion ($\lambda=0.3$) produces a distribution shifted toward the strongest effects, surpassing the best monolingual mask.



(a) ESS Distribution, Deactivation

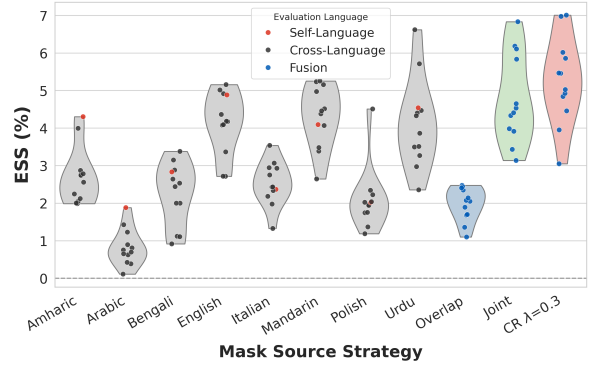


Figure 10: ESS distributions across monolingual and fusion masks under (a) deactivation and (b) steering.

LALM	Group	Monolingual ESN Masks								Fusion MLEN Masks				
		English	Mandarin	Amharic	Arabic	Bengali	Italian	Polish	Urdu	Overlap	Joint	CR ($\lambda=0.3$)		
Audio Flamingo 3	$\mathcal{L}_{id}$	English	-7.43	-5.17	-3.30	-0.63	-3.77	-1.67	-4.70	-4.67	-8.03	-7.60	-9.27	
		Mandarin	-8.97	-5.97	-4.33	-0.17	-3.87	-2.30	-6.03	-8.70	-14.23	-11.87	-12.53	
		Amharic	-3.90	-2.40	-4.50	-2.27	-2.83	-1.80	-4.60	-5.33	-12.30	-8.90	-9.70	
		Arabic	-6.30	-2.97	-3.53	-3.43	-2.90	-1.23	-5.33	-5.63	-7.37	-9.97	-10.03	
		Bengali	-6.70	-4.60	-5.67	-1.63	-3.67	-3.23	-5.57	-8.13	-14.33	-11.17	-11.97	
		Italian	-4.37	-4.20	-2.83	-2.63	-2.33	-2.70	-4.23	-4.63	-8.07	-8.40	-8.40	
		Polish	-2.20	-1.73	-2.10	-0.43	-1.47	-0.77	-1.87	-2.93	-3.90	-4.27	-4.77	
		Urdu	-7.50	-4.43	-4.53	-0.23	-2.50	-2.50	-5.27	-9.27	-13.00	-11.47	-12.70	
	$\mathcal{L}_{held}$	French	-6.46	-3.89	-4.13	-2.01	-3.12	-1.77	-5.59	-4.10	-7.64	-11.49	-10.83	
		German	-2.36	-5.76	-1.38	-1.54	-4.13	-1.85	-3.76	-1.99	-1.24	-6.27	-5.50	
		Persian	-6.32	-5.44	-3.00	-1.82	-4.76	-1.59	-3.22	-4.84	-11.95	-11.53	-13.24	
		Russian	-7.43	-4.27	-4.73	0.00	-1.73	-1.37	-4.73	-7.53	-14.60	-9.00	-10.13	
	Kimi-Audio	$\mathcal{L}_{id}$	English	-13.17	-10.27	-5.60	-2.10	-9.47	-9.33	-5.00	-10.27	-4.70	-13.07	-13.97
			Mandarin	-10.87	-11.40	-4.53	-2.40	-8.97	-7.83	-3.90	-11.87	-4.93	-13.70	-14.00
			Amharic	-11.70	-11.90	-11.30	-5.60	-14.60	-15.13	-7.70	-13.43	-9.00	-19.43	-19.17
			Arabic	-16.00	-15.23	-9.61	-6.51	-13.37	-16.90	-7.53	-15.93	-9.40	-19.20	-19.63
Bengali			-16.87	-15.80	-8.30	-4.50	-13.40	-14.87	-4.43	-15.90	-9.70	-23.43	-22.13	
Italian			-12.07	-11.43	-6.10	-3.70	-10.50	-13.63	-2.70	-11.57	-6.40	-14.93	-15.93	
Polish			-7.90	-6.20	-4.37	-2.87	-5.80	-8.57	-2.73	-8.93	-4.67	-9.93	-10.17	
Urdu			-15.43	-14.13	-7.97	-2.93	-10.37	-16.17	-4.77	-15.90	-6.67	-19.73	-18.97	
$\mathcal{L}_{held}$		French	-8.51	-7.19	-5.56	-4.58	-7.50	-11.22	-2.50	-12.19	-6.53	-15.00	-16.60	
		German	-9.94	-12.57	-2.48	-2.80	-7.91	-12.28	-3.97	-13.65	-6.05	-15.07	-17.73	
		Persian	-21.22	-19.68	-8.85	-1.99	-15.13	-20.62	-5.36	-21.27	-11.01	-25.23	-28.48	
		Russian	-11.53	-10.73	-6.40	-5.13	-9.40	-10.43	-2.03	-12.97	-7.07	-15.00	-14.43	
MiniCPM -o-4.5		$\mathcal{L}_{id}$	English	-5.80	-5.93	0.07	-0.33	-3.53	-1.70	-2.47	-4.80	-4.43	-4.73	-5.70
			Mandarin	-3.30	-3.13	-0.53	0.03	-1.23	-1.03	-1.27	-2.83	-3.07	-2.43	-3.67
			Amharic	-9.23	-11.20	-3.50	-1.10	-5.53	-3.07	-9.47	-12.70	-9.97	-12.23	-14.00
			Arabic	-2.27	-1.87	-0.33	-0.33	-1.30	-0.67	-0.70	-1.73	-1.37	-1.57	-1.80
	Bengali		-10.80	-13.13	-6.00	-0.57	-3.30	-6.70	-0.90	-11.60	-5.30	-9.80	-13.50	
	Italian		-6.20	-5.80	-1.43	-1.80	-4.27	-1.60	-4.63	-6.33	-5.70	-6.00	-6.63	
	Polish		-1.47	-1.43	-0.30	-0.30	-0.70	-0.50	-1.00	-1.20	-1.10	-1.03	-1.37	
	Urdu		-9.07	-11.30	-4.53	-0.17	-3.20	-5.60	-0.97	-8.60	-4.17	-7.13	-11.67	
	$\mathcal{L}_{held}$	French	-9.72	-12.01	-4.79	-1.39	-3.09	-6.74	-4.37	-11.49	-6.11	-10.03	-13.16	
		German	-5.63	-6.37	-1.07	-1.04	-3.73	-2.95	-2.47	-4.98	-3.60	-4.66	-5.77	
		Persian	-9.50	-11.60	-5.10	-0.07	-4.07	-4.50	-2.90	-10.96	-7.23	-9.93	-11.77	
		Russian	-6.60	-7.90	-2.70	-0.47	-2.53	-3.47	-1.57	-5.73	-4.47	-5.60	-7.33	
	Qwen2.5-Omni-7B	$\mathcal{L}_{id}$	English	-10.47	-8.73	-5.57	-2.90	-5.63	-0.30	0.20	-3.03	0.37	-5.20	-8.40
			Mandarin	-11.57	-10.03	-5.00	0.43	-6.93	-2.77	-0.77	-1.23	-2.07	-4.87	-10.40
			Amharic	-5.03	-5.20	-8.13	-0.53	-0.97	-2.70	-1.13	-3.03	2.13	-7.30	-5.67
			Arabic	-7.97	-6.70	-5.87	-2.20	-1.50	-1.43	-1.97	-0.57	2.53	-6.03	-6.07
Bengali			-5.47	-6.07	-7.50	0.37	-2.17	-0.83	0.50	-3.23	1.40	-5.17	-4.10	
Italian			-3.97	-2.77	-6.60	-3.37	-3.47	-3.07	2.27	-2.73	0.27	-5.53	-5.43	
Polish			-4.67	-4.50	-8.00	2.03	-0.90	-3.33	-0.27	2.30	1.87	-5.77	-4.23	
Urdu			-7.60	-6.23	-6.97	-0.47	-3.27	-4.53	-1.10	-4.23	-1.10	-7.43	-9.13	
$\mathcal{L}_{held}$		French	-7.43	-5.66	-5.80	0.00	-2.57	-3.99	-0.52	-3.99	1.63	-6.49	-5.59	
		German	-5.66	-5.72	-7.83	-0.58	-3.05	-4.55	1.88	-5.15	-2.51	-6.34	-10.75	
		Persian	-5.41	-5.12	-4.41	0.56	-1.72	-3.56	-0.19	-3.73	2.26	-6.02	-7.32	
		Russian	-8.60	-7.13	-6.03	-2.23	-6.17	-3.17	-0.03	-2.47	0.20	-6.50	-7.77	

Table 8: Per-language deactivation effects. More negative values indicate stronger causal necessity of deactivated neurons.

LALM	Group	Lang. Set	Monolingual ESN Masks								Fusion MLEN Masks		
			English	Mandarin	Amharic	Arabic	Bengali	Italian	Polish	Urdu	Overlap	Joint	CR ($\lambda=0.3$)
Audio Flamingo 3	$\mathcal{L}_{id}$	English	4.47	3.53	1.13	0.10	1.27	1.37	2.83	3.33	2.37	3.77	4.60
		Mandarin	4.20	2.57	1.93	-0.03	1.67	1.23	2.37	3.43	2.63	3.87	4.50
		Amharic	4.00	1.13	3.50	0.63	1.60	1.17	2.27	5.03	2.47	5.53	6.03
		Arabic	5.10	1.70	2.43	1.40	1.60	0.77	5.43	4.50	3.30	5.83	5.40
		Bengali	4.30	2.37	3.27	0.27	2.73	0.80	2.17	4.57	2.87	6.30	6.60
		Italian	1.70	1.80	0.57	0.67	1.27	0.63	1.97	2.03	1.27	3.17	3.10
		Polish	3.00	1.40	1.27	-0.03	0.63	0.47	3.13	2.40	1.37	3.53	3.30
		Urdu	3.00	2.40	1.00	0.60	1.33	0.83	2.40	3.47	1.50	4.00	3.90
	$\mathcal{L}_{held}$	French	2.95	1.88	1.87	0.66	2.36	2.36	2.12	3.09	2.26	5.10	5.28
		German	1.12	1.48	0.68	-0.14	0.57	0.65	1.00	0.52	-0.22	1.75	1.84
		Persian	2.99	2.36	1.93	0.32	1.55	1.48	1.10	2.15	1.99	4.32	4.69
		Russian	3.80	2.43	2.13	0.27	1.13	0.27	2.70	3.23	2.00	4.00	4.20
Kimi-Audio	$\mathcal{L}_{id}$	English	5.70	4.83	2.60	0.80	4.10	4.23	2.10	4.70	2.70	5.43	5.63
		Mandarin	5.27	5.67	2.13	2.37	3.83	3.93	1.93	4.73	2.37	6.33	7.17
		Amharic	6.30	6.57	5.37	4.07	6.53	6.90	3.30	8.27	5.30	9.40	8.70
		Arabic	8.53	8.33	4.73	4.13	7.97	7.37	4.50	7.93	5.07	10.97	10.67
		Bengali	5.73	4.87	4.63	2.13	5.13	6.13	1.67	6.20	3.97	8.07	8.03
		Italian	4.47	3.77	1.90	1.30	4.60	5.30	0.57	4.73	2.50	7.30	7.17
		Polish	3.17	2.70	1.83	1.10	2.43	2.77	0.83	3.73	1.97	4.23	4.60
		Urdu	3.73	4.77	2.57	1.03	3.80	4.03	2.60	4.37	2.40	6.23	6.07
	$\mathcal{L}_{held}$	French	4.51	2.88	1.74	1.81	2.71	3.19	0.87	4.34	2.92	5.03	5.56
		German	2.48	3.59	1.18	2.48	2.52	3.06	1.88	3.83	2.92	4.35	4.34
		Persian	6.74	6.43	3.55	1.43	7.08	5.71	2.76	6.08	4.11	9.03	8.13
		Russian	5.07	5.23	3.23	2.13	4.30	4.23	1.30	6.33	3.63	6.73	7.33
MiniCPM -o-4.5	$\mathcal{L}_{id}$	English	3.83	3.60	0.50	-0.03	2.20	1.07	2.07	3.03	2.43	4.27	3.67
		Mandarin	2.40	3.33	0.33	0.23	1.93	1.37	2.00	2.83	1.60	2.90	3.77
		Amharic	6.30	7.77	4.00	0.97	2.40	3.80	3.43	9.07	3.70	5.00	10.07
		Arabic	2.20	3.23	1.13	0.13	1.60	1.27	2.73	3.93	1.83	3.37	4.07
		Bengali	6.80	8.53	4.27	-0.03	3.03	4.27	2.23	8.53	3.87	7.40	10.33
		Italian	4.63	5.30	1.43	0.93	2.93	1.97	5.77	5.50	4.33	4.83	5.90
		Polish	1.63	2.13	0.60	0.20	1.37	0.53	3.13	2.43	1.73	2.83	2.17
		Urdu	5.67	7.30	3.57	-0.30	1.37	3.43	1.10	6.03	1.20	3.73	7.77
	$\mathcal{L}_{held}$	French	5.10	5.45	3.75	-0.17	1.08	3.47	1.74	6.60	2.78	2.81	7.08
		German	4.65	5.65	1.74	0.48	2.45	2.32	3.28	4.68	2.65	3.84	5.60
		Persian	5.39	6.07	2.87	-0.37	2.63	2.03	1.87	5.93	3.73	4.97	8.16
		Russian	3.63	4.77	2.23	-0.30	1.10	2.80	0.80	3.77	1.80	3.90	5.07
Qwen2.5- Omni-7B	$\mathcal{L}_{id}$	English	5.53	4.50	3.33	1.67	2.57	0.80	0.00	2.10	0.07	3.87	5.47
		Mandarin	4.83	4.03	3.50	1.07	2.33	2.33	0.30	0.33	1.70	2.83	4.27
		Amharic	3.07	4.23	3.53	0.03	1.00	1.60	0.00	2.57	-1.57	4.80	3.10
		Arabic	4.23	3.83	2.23	0.57	1.43	1.03	2.20	0.60	-0.80	4.27	3.30
		Bengali	3.80	3.43	2.97	-0.30	0.43	0.37	-0.43	1.97	-0.93	5.57	3.07
		Italian	2.67	1.70	4.47	0.33	1.77	0.93	-0.30	0.80	0.10	2.83	1.67
		Polish	3.07	3.73	4.30	-1.07	-0.77	1.23	-0.30	0.33	-0.67	3.13	2.13
		Urdu	4.33	4.00	2.47	-0.20	1.50	0.77	0.47	2.93	0.33	3.67	4.10
	$\mathcal{L}_{held}$	French	3.75	2.64	3.51	0.31	-1.67	2.12	0.28	2.26	-1.22	2.71	2.19
		German	2.62	4.40	3.89	0.73	-1.11	2.04	-1.98	3.05	1.46	2.61	4.04
		Persian	2.33	4.68	1.87	-0.03	2.25	1.72	1.02	1.87	-0.20	5.02	3.09
		Russian	3.87	4.17	2.77	-0.40	1.47	1.77	0.63	0.93	1.13	3.97	5.27

Table 9: Per-language steering effects ($\alpha=0.5$). More positive values indicate stronger causal sufficiency of amplified neurons.